https://openstax.org/books/contemporary-mathematics/pages/7-11-expected-value

That definition is a little vague: How many is "a large number?" In practice, it depends on the experiment; the number has to be large enough that every outcome would be expected to appear at least a few times. For example, if we're talking about ***rolling a standard 6-sided die and we note the number showing, a few dozen replications should be enough that the mean would be representative***. Since the probability of each outcome is 1/6, we would expect to see each outcome about 8 times over the course of 48 replications. **However, if we're talking about the Powerball lottery, where the probability of winning the jackpot is about (1/292,000,000), we would need several *billion* replications to ensure that every outcome appears a few times**. Luckily, we can find the theoretical expected value before we even run the experiment the first time.

6-Face Die

$\rightarrow 6 \times 8 = 48 \leftarrow$ Tosses

$$E(X) = \frac{1}{6}(1) + \frac{1}{6}(2) + \dots + \frac{1}{6}(6) = 3.5$$

$$\boxed{E(X) = x_i f(x_i)}$$

$$Var(X) = \sum_{i=1}^{6} (x_i - \bar{x})^2 f(x_i)$$

$$= \sum_{i=1}^{6} (x_i - E(X))^2 f(x_i)$$

$$E(X) = \mu = 3.5$$

$$Var(X) = (1-3.5)^2\left(\frac{1}{6}\right) +$$

$$+ (2-3.5)^2\left(\frac{1}{6}\right)$$

$$\vdots$$

$$+ (6-3.5)^2\left(\frac{1}{6}\right)$$

$$Var(X) = .917$$

$$SD(X) = \sqrt{Var(X)} = 1.71$$

Roll 2 Die

Die 1

Die 2	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$N = 36$

$$f(7) = Prob(X_i = 7) = \frac{6}{36} = \frac{1}{6}$$

$\boxed{E(X) = 7}$ $\boxed{Var(X) = 5.83}$

$SD(X) = 2.41$

CONTINUOUS PROBABILITY DISTRIBUTIONS

⊛ NORMAL DISTRIBUTION

$\longrightarrow \sigma^2 \longrightarrow$ z - test (known)

$\hat{\sigma}^2 \longrightarrow$ t - test

↳ estimates since we do NOT know σ^2, etc

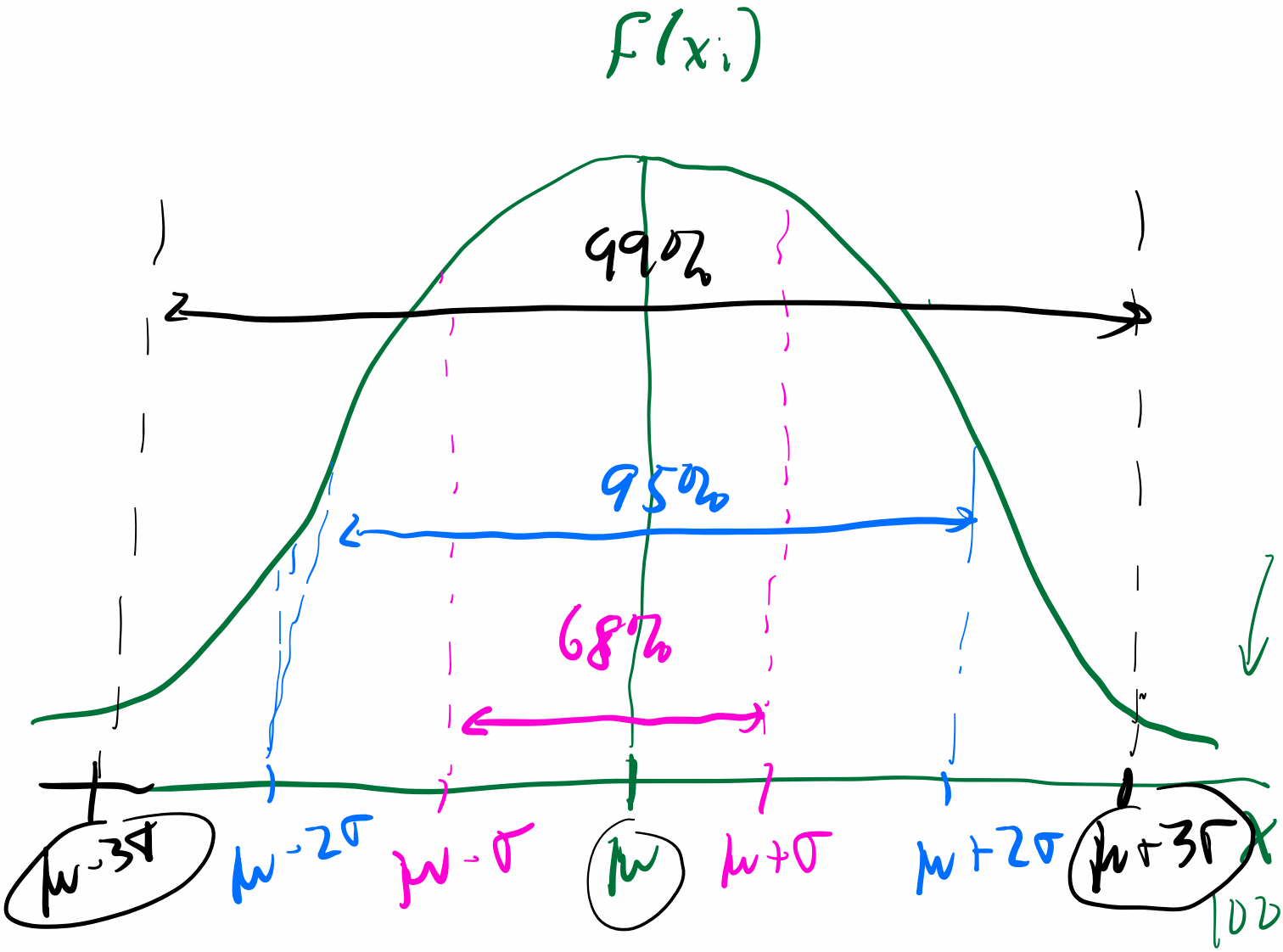
The NORMAL DISTRIBUTION

Terms

Suppose X = Random Variable

$$\boxed{X \sim N(\mu, \sigma^2)}$$

1. "$\sim$" Distribution of RV
2. $N \rightarrow$ NORMAL DISTRIBUTION
3. $\mu \rightarrow$ POPULATION MEAN
4. σ^2 - population variance



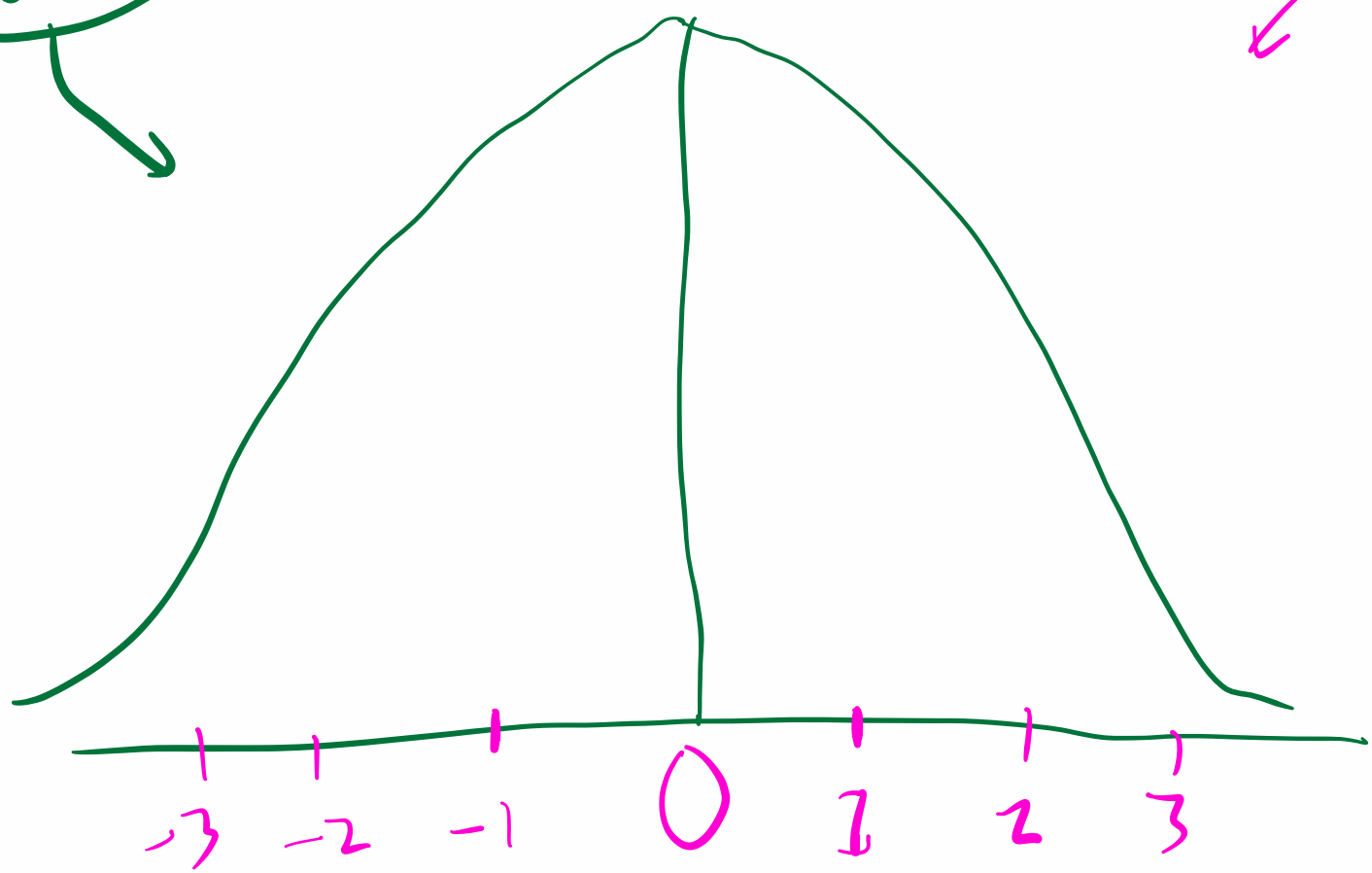
(I) $(\mu \pm \sigma) \rightarrow$ 68% of area under curve

$(\mu \pm 2\sigma) \rightarrow$ 95%

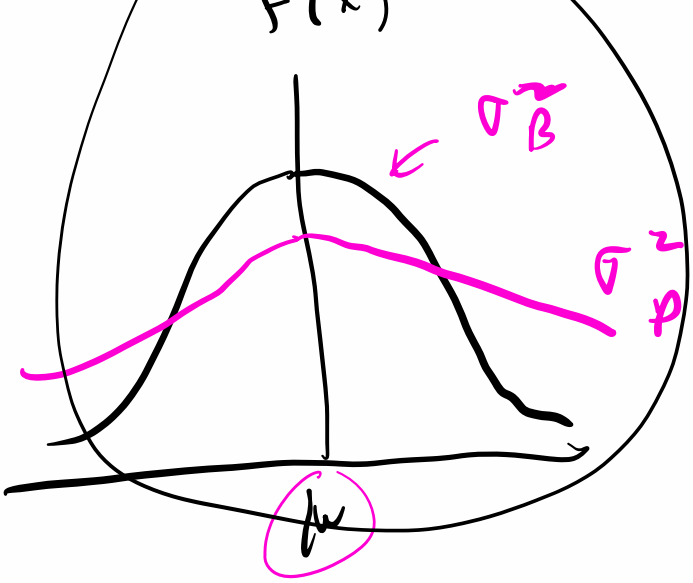
$(\mu \pm 3\sigma) \rightarrow$ 99%

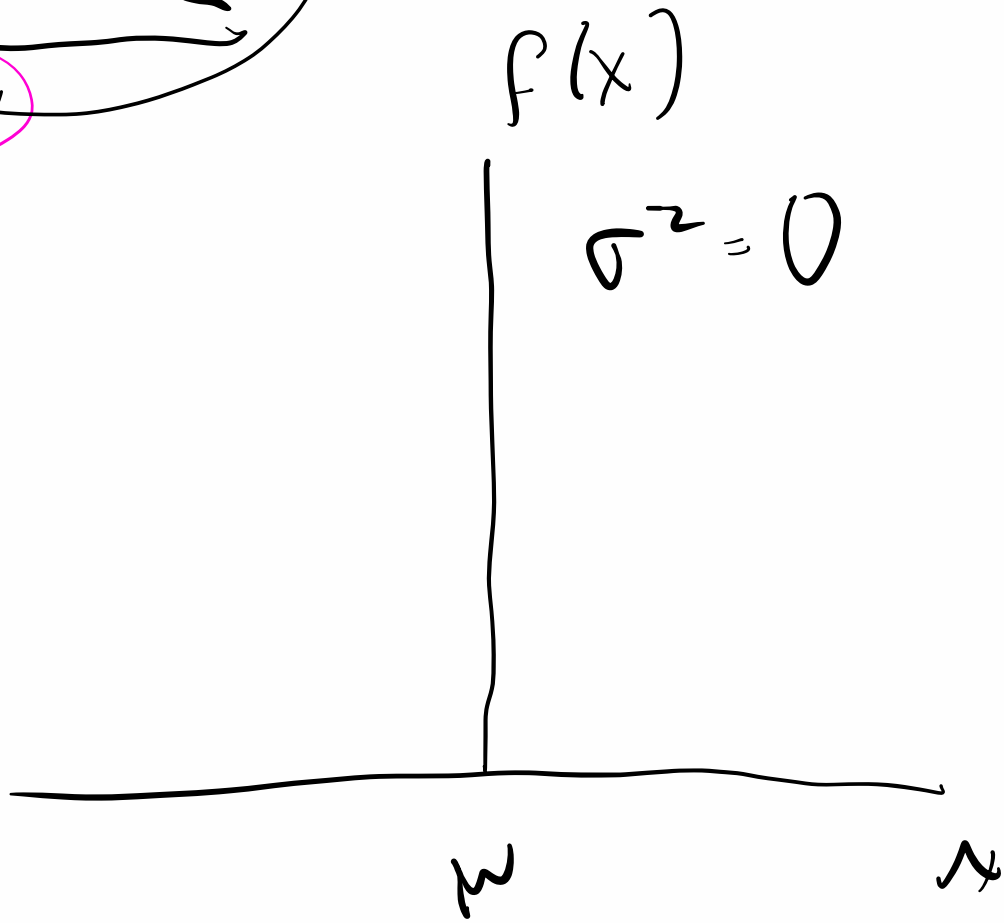
$$z = \frac{x - \mu}{\sigma}$$

z - TABLE



$$Z \sim N(0, 1)$$


f(x)
σ_B^2
σ_P^2
μ


f(x)
$\sigma^2 = 0$
μ
x